

P3 Advanced practical skills

Practical · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Plan to vary one independent variable while controlling others. Calibrate instruments, record raw data with uncertainties and consistent precision, plot a graph with labelled axes and appropriate scale, then calculate gradient using a large triangle. Repeat measurements to identify anomalies.

Worked example: follow the reasoning

If length is measured as 0.500 ± 0.001 m, percentage uncertainty is 0.2%; record to nearest millimetre if that is instrument resolution.

Quick check · multiple choice

A In a fair pendulum trial, which variable should remain controlled? [1]

- A String length if studying length
- B Room temperature only
- C Number of pages in notes
- D Date of experiment

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A pendulum experiment records times for 10 oscillations. Explain why timing several oscillations improves the estimate of period, and name two controlled variables. [3]

Final answer: _____

Answers and explanation

Quick check: A. String length if studying length

Question 1 [3 marks]

Longer total interval reduces reaction-time percentage error [1]; control amplitude and bob/string setup or release method [2].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.